Probabilistic Robotics Exam Sample
Dec 17 2021

December 16, 2021

1 Question: Data Association

Illustrate the problem of data association, and present potential solutions in the Gaussian case. Discuss the heuristics that can be used to quickly compute an approximated solution.

2 Solutuon Question 1

Slides 11. Problem definition : Slide number 4 and 5. Gaussian Case : 14 to 25. Heuristics 26 to 31.

3 Exercise: Indoor Localization

Scenario: An indoor GPS system consists of a set of beacons at known locations in the environment. These beacons transmit a data packet at periodic intervals. A packet set by the i^{th} beacon contains the following information:

- time when the packet was transmitted
- location $\mathbf{p}_i$ of the beacon

The clocks of the beacons are synchronized *between them*. A robot equipped with a receiver that can decode this packets, and timestamp their arrival. The packets are transmitted through UWB, and so they traverse the environment at the speed of light. If the clocks of the beacons and of the robot would be synchronized, the travel time of the message is proportional to the distance between transmitter and receiver.

The clock on the robot is *not synchronized* with the one of the beacons. The robot has a differential drive kinematics, moves on a plane, and is controlled issuing a desired velocity to the left and the right wheels (radians per second). The parameters of the platform b, k_r, k_l are unknown. The control interval is ΔT, and the velocity is assumed to be constant within the interval. The controls are affected by an additive gaussian noise $\mathbf{n}_u \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_u)$. The receiver is at the center of the robot. In addition to the control noise, the system dynamics is subject to an addittive white process noise $\mathbf{n}_x \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_x)$ so that:

$$\mathbf{x}_t = f(\mathbf{x}_{t-1}, \mathbf{u}_{t-1}) + \mathbf{n}_x \tag{1}$$

3.1 Question 1

Model the dynamic system by identifying:

- State
- Controls
- Measurements
- Transition Function
- Observation Function

3.2 Question 2

After having identified the system, approach the problem of estimating the robot position with a filter at your choice. Report the filter equations, for the specific problem and a sketch of filter implementation. If Jacobians are involved report the sizes.

3.3 Question 3

Discuss the implementation of the problem with filters different from the one chosen in Question 2.

4 Solution Question 1

- The state of the system $\mathbf{x}$ include:
  - Robot pose $\mathbf{x}_R \in SE(2)$, parametrized on $\mathbb{R}^3$
  - Intrinsic parameters $\mathbf{x}_K = (k_r, k_l, b) \in \mathbb{R}^3$
  - Time delay between the receiver and the beacons $\mathbf{x}_\tau$

  And so $\mathbf{x} = (\mathbf{x}_R, \mathbf{x}_K, \mathbf{x}_\tau)^T \in \mathbb{R}^7$.
- Control : $\mathbf{u} = (u_l, u_r) \in \mathbb{R}^2$

- Measurements : Let t_{tx} be clock time of the beacon when the packet was transmitted and let t_{rx} be the clock time of the robot when the packet was received. We can combine this two informations as:

$$\mathbf{z} = t_{\text{rx}} - t_{\text{tx}} \tag{2}$$

- Transition model : The two angular velocities of the left and right wheel are (u_l, u_r).

  We report here a simplified model that computes the robot relative displacement by integrating the wheel motion within the interval, and then uses this estimate to update the previous position:

$$\begin{aligned}
\Delta_{x,t} &\simeq \frac{u_r \cdot k_r + u_l \cdot k_l}{2} \cdot \Delta T \\
\Delta_{y,t} &\simeq 0 \\
\Delta_{\theta,t} &= \frac{u_r \cdot k_r - u_l \cdot k_l}{b} \cdot \Delta T
\end{aligned}$$

  The complete transition model is then:

$$\begin{aligned}
\mathbf{x}_{R,t} &= \mathbf{x}_{R,t-1} + \begin{pmatrix} \cos(\theta_t) \cdot \Delta_{x,t} \\ \sin(\theta_t) \cdot \Delta_{x,t} \\ \boldsymbol{\Delta}_{\theta,t} \end{pmatrix} && (3) \\
\mathbf{x}_{K,t} &= \mathbf{x}_{K,t-1} && (4) \\
\mathbf{x}_{\tau,t} &= \mathbf{x}_{\tau,t-1} && (5) \\
& && (6)
\end{aligned}$$

- Measurement function :

$$\mathbf{h}(\mathbf{x}) = \frac{\|\mathbf{x}_R^{-1} \cdot \mathbf{p}_i\|}{c_0} - \mathbf{x}_\tau \tag{7}$$

  Where c_0 is the speed of light. The first term in (7), represents the travel time of the packet given the current estimate of the robot pose, while $\mathbf{x}_\tau$ compensate for the clock difference between transmitter and receiver.

4.1 Solution Question 2 and 3

4.1.1 EKF

- Initial State and control distributions:

$$\begin{aligned}
p(\mathbf{x}_{0,0}) &\sim \mathcal{N}(\mathbf{x}; \mu_{0,0}, \boldsymbol{\Sigma}_{0,0}) \\
p(\mathbf{u}) &\sim \mathbf{u}_t + \mathbf{n}_u
\end{aligned}$$

- Jacobian of the transition model with respect to the state:

$$\mathbf{A}_t \in \mathbb{R}^{7\times 7}$$

- Jacobian of the transition model with respect to the controls:

$$\mathbf{B}_t \in \mathbb{R}^{7\times 2}$$

- Jacobian of the measurement function:

$$\mathbf{C}_t \in \mathbb{R}^{1\times 7}$$

- Transition :

$$\mu_{t|t-1} = \mathbf{A}_t \mu_{t-1|t-1} + \mathbf{B}_t \mathbf{u}_{t-1}$$
$$\mathbf{\Sigma}_{t|t-1} = \mathbf{A}_t \mathbf{\Sigma}_{t-1|t-1} \mathbf{A}_t^T + \mathbf{B}_t \mathbf{\Sigma}_u \mathbf{B}_t^T + \mathbf{\Sigma}_x$$

- Correction :

$$\mu_z = \mathbf{C}_t \mu_{t|t-1}$$
$$\mathbf{K}_t = \mathbf{\Sigma}_{t|t-1} \mathbf{C}_t^T \left(\mathbf{\Sigma}_z + \mathbf{C}_t \mathbf{\Sigma}_{t|t-1} \mathbf{C}_t^T \right)^{-1}$$
$$\mu_{t|t} = \mu_{t|t-1} + \mathbf{K}_t \left(\mathbf{z}_t - \mu_z \right)$$
$$\mathbf{\Sigma}_{t|t} = \left(\mathbf{I} - \mathbf{K}_t \mathbf{C}_t \right) \mathbf{\Sigma}_{t|t-1}$$

4.1.2 UKF

- Initial State and control distributions:

$$p(\mathbf{x}_{0,0}) \sim \mathcal{N}(\mathbf{x}; \mu_{0,0}, \mathbf{\Sigma}_{0,0})$$
$$p(\mathbf{u}) \sim \mathbf{u}_t + \mathbf{n}_u$$

- Complete distribution:

$$p(\mathbf{x}_{t-1|t-1}, \mathbf{u}_{t-1}) \sim \mathcal{N} \left(\begin{pmatrix} \mu_{t-1|t-1} \\ \mathbf{u}_{t-1} \end{pmatrix}, \begin{bmatrix} \mathbf{\Sigma}_{t-1|t-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_u \end{bmatrix} \right) \tag{8}$$

- Compute the 19 sigma points $\mathcal{X}_{t-1|t-1}$ from the complete distribution.
- Transition: apply (3) to all the sigma points to obtain $\mathcal{X}_{t|t-1}$
- Correction:
  - Compute mean of the state and predicted observation:

$$\mathbf{z}_t^{(i)} = \mathbf{h}(\mathcal{X}_{t|t-1}^{(i)})$$
$$\mu_z = \sum_i w_m^{(i)} \mathbf{z}_t^{(i)}$$
$$\mu_{t|t-1} = \sum_i w_m^{(i)} \mathbf{x}_{t|t-1}^{(i)}$$

  - Reconstruct the required blocks of the covariance:

$$\mathbf{\Sigma}_{x,z} = \sum_i w_c^{(i)} (\mathbf{x}_{t|t-1}^{(i)} - \mu_{t|t-1})(\mathbf{z}_t^{(i)} - \mu_z)^T$$
$$\mathbf{\Sigma}_{z,z} = \sum_i w_c^{(i)} (\mathbf{z}_t^{(i)} - \mu_z)(\mathbf{z}_t^{(i)} - \mu_z)^T$$

  - Apply the Kalman equation:

$$\mu_{t|t} = \mu_{t|t-1} + \mathbf{\Sigma}_{x,z} \left(\mathbf{\Sigma}_{z|x} + \mathbf{\Sigma}_{z,z} \right)^{-1} (\mathbf{z}_t - \mu_z)$$
$$\mathbf{\Sigma}_{t|t} = (\mathbf{\Sigma}_{t|t-1} + \mathbf{\Sigma}_x) - \mathbf{\Sigma}_{x,z} \left(\mathbf{\Sigma}_{z|x} + \mathbf{\Sigma}_{z,z} \right)^{-1} \mathbf{\Sigma}_{z,x}$$

4.1.3 PF

- Initial particle set:

$$\mathcal{P}_0 = \{(\mathbf{x}_0^{(i)}, w_0^{(i)}) \mid i \in [1, 2, \dots N]\} \tag{9}$$

- Control noise distribution:

$$p(\mathbf{u}) \sim \mathbf{u}_t + \mathbf{n}_u \tag{10}$$

- Transition : apply (3) to each element of $\mathcal{P}_{t-1}$. For each particle draw a different sample from the control and process noise, $(u_{t-1}^{(i)}, \delta_x^{(i)})$, so that:

$$\mathbf{x}_t^{(i)} = f(\mathbf{x}_{t-1}^{(i)}, u_{t-1}^{(i)}) + \delta_x^{(i)} \tag{11}$$

- Correction : As likelihood function $p(\mathbf{z}_t|\mathbf{x}_t^{(i)})$ i chose :

$$p(\mathbf{z}_t|\mathbf{x}_t^{(i)}) = \frac{1}{2}\, e^{-|\mathbf{h}(\mathbf{x}_t^{(i)})-\mathbf{z}_t|} \tag{12}$$

  With $\mathbf{h}(\mathbf{x})$ defined in (7).
  The correction step involve just the particle weights as:

$$w_t^{(i)} = w_{t-1}^{(i)} \cdot p(\mathbf{z}_t|\mathbf{x}_t^{(i)}) \tag{13}$$

- Resample

5 Exercise: Map The Stars

Scenario: a robot moves on the surface of a planet that can be approximated by a sphere.

The robot is equipped with a (3D) odometer, that reports the relative motion between subsequent poses of the robot on the surface. The robot has a spherical camera pointed upwards, that observes the sky and sees the stars. In this scenario we assume that the camera can identify the stars.

The stars can be seen as infinitely far from the robot, thus we can observe only the azimuth and the elevation of each star.

5.1 Question 1

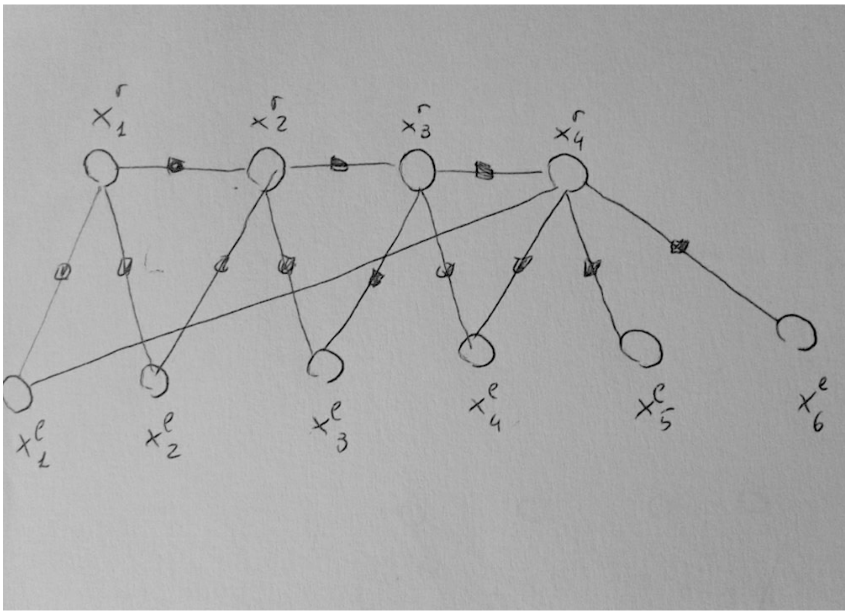
Formulate a least squares problem that, computes the trajecory of the robot and the position of the stars as azimuth and elevation with respect to the initial position of the robot.

Define:

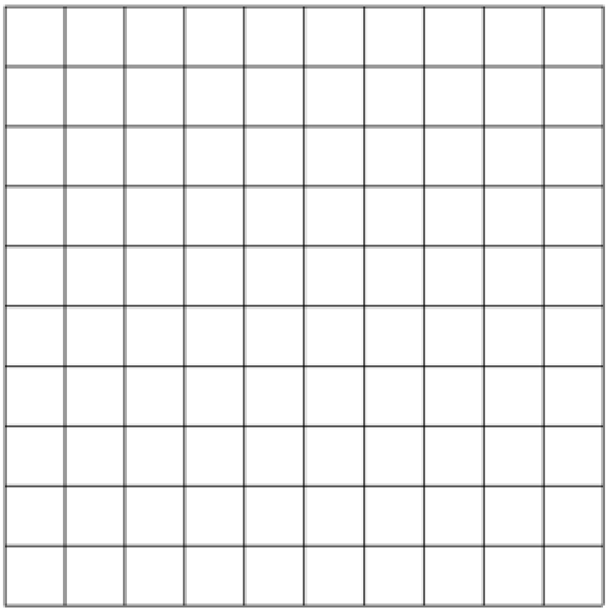
- State: Domains, parameterization of the increments, $\boxplus$ if needed.
- Measurements: Domains, $\boxminus$ if needed.
- Prediction function
- Error function
- The size of the Jacobians

5.2 Question 2

Consider the simple problem illustrated in the following factor graph:



where $\mathbf{x}^l$ denotes a star and $\mathbf{x}^r$ denotes a robot pose. Show the block structure of the H matrix, by filling the grid provided here:



5.3 Question 3

In case the data association would not be known how would you approach the problem?. Consider the case of sequential observations, and assume the stars are not too close to each other.

5.4 Solution Question 1

For the fact that you have an odometer a pose-pose measurement about the relative motion of the robot is always available. To include that in the system you have to define another error which handle pose-pose constraints (which you can find in slides 27 (multi-pose registration), from 20 to 26). Here i will report prediction function, error and Jacobian just for the star measurements

5.4.1 Preliminaries

Lets define a function that map azimuth and elevation $\mathbf{a} = (\theta, \phi)$ to a 3D rotation matrix :

$$\mathcal{R}(\mathbf{a}) = R_z(\theta) \cdot R_y(\phi) \tag{14}$$

Where $R_z(\theta), R_y(\phi)$ are elementary rotations around the z-axis and the y-axis. Further, let's define another function which, given a unit vector $\mathbf{v} = (v_x, v_y, v_z)^T$, computes the corresponding azimuth and elevation:

$$\Gamma(\mathbf{v}) = \begin{pmatrix} atan2(\frac{v_y}{v_x}) \\ atan2(\frac{v_z}{v_x^2 + v_y^2}) \end{pmatrix} \tag{15}$$

5.4.2 Methodology

I decide to represent each star as a unit vector which point to the estimated location of the star (the range is not observable). Those unit vector can be easily mapped in azimuth and elevation using the Γ function.
To represent the perturbation i use azimuth and elevation directly (not unit vector). To apply the perturbation, I first convert those angles in a rotation matrix and then apply it to the actual estimate of a star (box-plus). With this choice of the parameterization/box-plus I can guaranty that the result is still a unit vector. Of course other parameterization are possible.

- We have N poses, represented as 3D transformation matrices and M stars represented as 3D unit vectors, so $\mathbf{x} \in SE(3)^N \times S^{2M}$ (S^2 represent the unit sphere manifold)

- Euclidean parameterization:
  - Pose : $\Delta\mathbf{x}_{r,i} \in \mathbb{R}^6$
  - Star : $\Delta\mathbf{x}_{s,i} \in \mathbb{R}^2$ (only azimuth and elevation)

  So $\Delta\mathbf{x} \in \mathbb{R}^{6N+2M}$.

- Box-Plus :
  - Pose :
  $$\mathbf{x}_{r,i} \boxplus \Delta\mathbf{x}_{r,i} = v2t(\Delta\mathbf{x}_{r,i}) \cdot \mathbf{x}_{r,i} \tag{16}$$
  - Star :
  $$\mathbf{x}_{s,i} \boxplus \Delta\mathbf{x}_{s,i} = \mathcal{R}(\Delta\mathbf{x}_{s,i}) \cdot \mathbf{x}_{s,i} \tag{17}$$

- The measurements are represented by two angles, $\mathbf{z} \in \mathbb{R}^2$.
- Prediction about azimuth and elevation of star $\mathbf{x}_{s,i}$ as seen by pose $\mathbf{x}_{r,j} = [\mathbf{R}_{r,j}|\mathbf{t}_{r,j}]$:
  $$\hat{\mathbf{z}}_{i,j}(\mathbf{x}) = \Gamma(\mathbf{R}_{r,j}^T \cdot \mathbf{x}_{s,i}) \tag{18}$$

- Even though the measurement are represented as element of $\mathbb{R}^2$, we know that they represent angular quantities. As such, in order to properly compare predictions and observations, we have to define a suitable box-minus operator. My choice is:
  $$\hat{\mathbf{z}}(\mathbf{x}) \boxminus \mathbf{z} = diag(\mathbf{I} - \mathcal{R}(\mathbf{z})^T \cdot \mathcal{R}(\hat{\mathbf{z}}(\mathbf{x}))) \tag{19}$$

  Alternatives for the box-minus :
  - Euclidean minus on the angles with renormalization of the result
  - Chordal distance on the rotation matrices $\mathcal{R}(\mathbf{z})$ and $\mathcal{R}(\hat{\mathbf{z}}(\mathbf{x}))$
  - Euclidean minus on the unit vectors, in this case one have to convert the measurements to unit vectors

- Error and Jacobian:
  $$\mathbf{e}_{p,s}(\mathbf{x}) = \hat{\mathbf{z}}(\mathbf{x}) \boxminus \mathbf{z} \tag{20}$$
  $$\mathbf{J}_{p,s}(\mathbf{x}) \in \mathbb{R}^{3\times(6N+2M)} \tag{21}$$

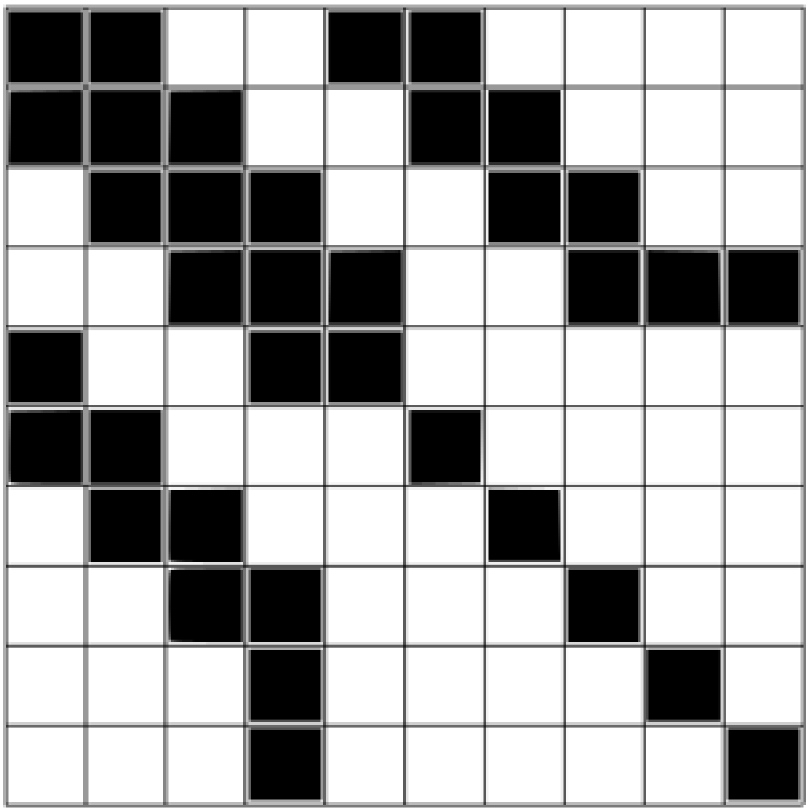
5.4.3 Structure of the Jacobians

- Jacobian of the pose-pose error function which involves pose i and pose j :
  $$\mathbf{J}_{pp}^{[i,j]}(\mathbf{x}) = (\cdots \mathbf{0}_{6\times6} \cdots \mathbf{J}_{p,i}(\mathbf{x}) \cdots \mathbf{0}_{6\times6} \cdots \mathbf{J}_{p,j}(\mathbf{x}) \cdots \mathbf{0}_{6\times2} \cdots \mathbf{0}_{6\times2} \cdots)$$
  Where $\mathbf{J}_{pp}^{[i,j]} \in \mathbb{R}^{6\times(6N+2M)}$ is the complete jacobian, while $\mathbf{J}_{p,i}(\mathbf{x}) \in \mathbb{R}^{6\times6}$ and $\mathbf{J}_{p,j}(\mathbf{x}) \in \mathbb{R}^{6\times6}$ are the non-zero blocks which correspond to pose i and pose j.

- Jacobian of the pose-star error function which involves pose i and star j :
  $$\mathbf{J}_{p,s}^{[i,j]}(\mathbf{x}) = (\cdots \mathbf{0}_{3\times6} \cdots \mathbf{J}_{p,i}(\mathbf{x}) \cdots \mathbf{0}_{3\times6} \cdots \mathbf{0}_{3\times6} \cdots \mathbf{J}_{s,j}(\mathbf{x}) \cdots \mathbf{0}_{3\times2} \cdots)$$
  Where $\mathbf{J}_{p,s}^{[i,j]} \in \mathbb{R}^{3\times(6N+2M)}$ is the complete jacobian, while $\mathbf{J}_{p,i}(\mathbf{x}) \in \mathbb{R}^{3\times6}$ and $\mathbf{J}_{s,j}(\mathbf{x}) \in \mathbb{R}^{3\times2}$ are the non-zero blocks which correspond to pose i and star j.

5.5 Solution Question 2

The correspondence between row/column index and variable is arbitrary, here i report the solution with a basic choice
Rows/Column order : $\{x_1^r, x_2^r, x_3^r, x_4^r, x_1^l, x_2^l, x_3^l, x_4^l, x_5^l, x_6^l\}$



5.6 Solution Question 3

Different solutions to this problem exist, here i will report my personal solution.

For the fact that the possible angles observed are in a fixed range $[-\pi, \pi]$, I will, at first, discretize the angle ranges at a fixed resolution r. Doing so, I can construct a grid where the rows represent the azimuth and the columns the elevation. Each cell of the grid contain either a 0 if no measurement (after discretization) corresponds to that value of azimuth and elevation, and a 1 otherwise. Under the assumption that the stars are far enough from each other, one cell correspond to exactly one star. Once the grid is constructed, i can compute a distance map on the top of it. By querying this data structure for each observation I can get the nearest star in the angular sense.